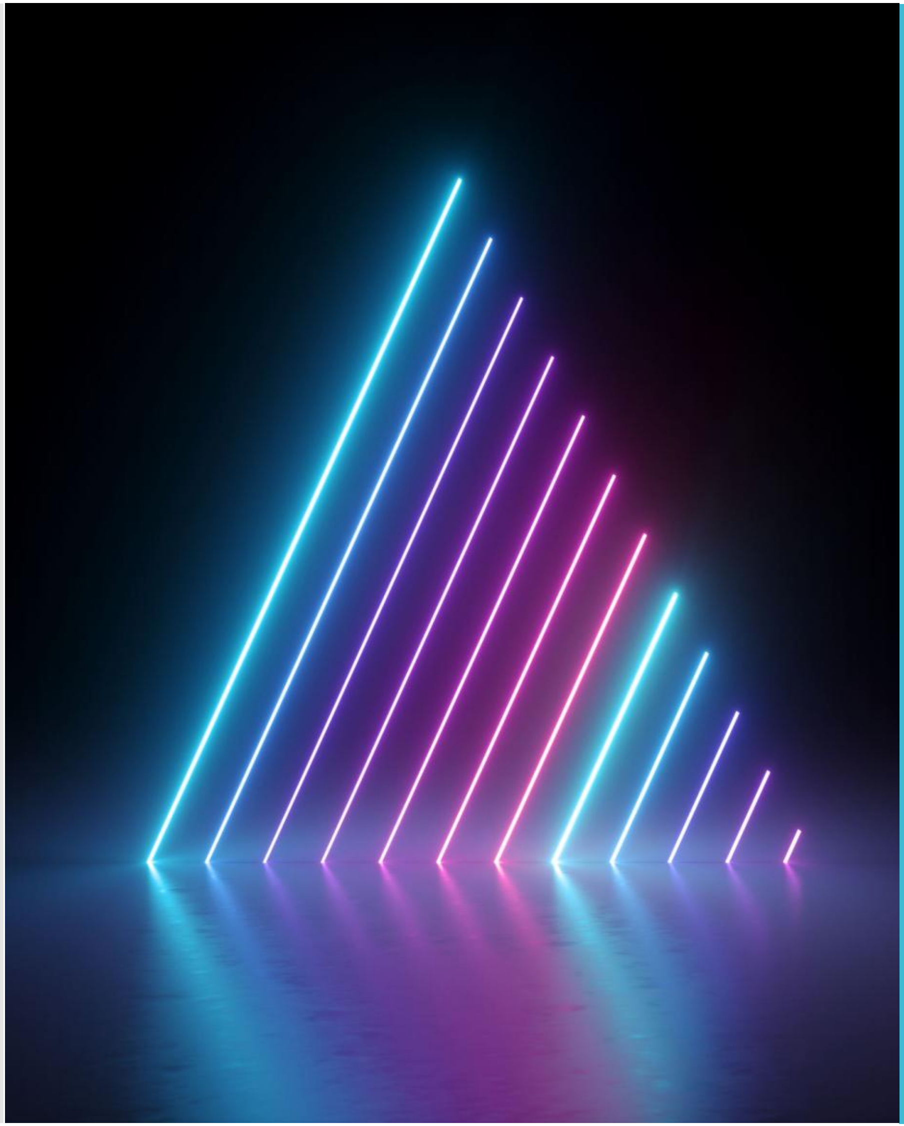


SWE 332
Software Architecture

Lecture 4

Quality attributes

SOFTWARE
ARCHITECTURE

Quality attributes

Also known as:

- ✓ Architecture characteristics.
- ✓ Non-functional requirements.

Types of requirements

- Requirements can be categorized as:
- **Functional requirements**: state what the system must do, how it must
  - behave or react to run-time stimuli.
- **Quality attribute requirements**. Annotate (qualify) functional requirements
  - Examples: Availability, modifiability, usability, security,...
  - Also called: non-functional requirements
- **Constraints**. A design decision that has already been made

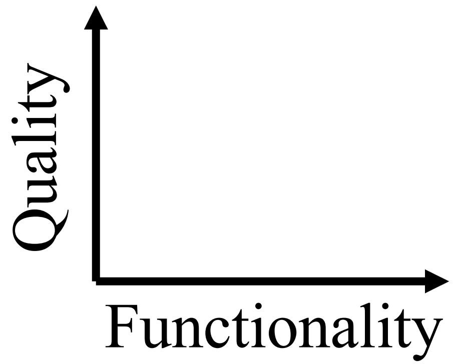
Functional requirements

Functionality = ability of the system to do the work for which it was intended.

Functionality has a strange relationship to architecture:

- It does not determine architecture;
  - If functionality were the only requirement, the system could exist as a single monolithic module with no internal structure at all

Functionality and quality attributes are orthogonal.



Quality attributes (QA)

Quality attributes annotate (qualify) the functionality.

If a functional requirement is "when the user presses the green buttom an options dialog appears"

- A performance QA could describe how quickly.
- An availability QA could describe this option could fail, or how often it will be repaired.
- A usability QA could describe how easy is to learn this function.

Quality attributes influence the architecture.

What is Quality?

Degree to which a system satisfies the stated and implied needs of its stakeholders, providing value

- Degree (not Boolean)
- Quality = Fitness for purpose (*stakeholders* needs)
- Conform to requirements (stated and implied)
- Providing value

Several definitions of quality

`https://en.wikipedia.org/wiki/Software_quality`

Quality attributes and trade-offs

QAs are all good

...but value depends on the project & stakeholder

- *"Best quality"...for what?, for whom?*

QAs are not independent

- Some qualities can conflict
- What matters most?
  - Example: A very secure system can be less usable

There is always a price

- What is your budget?

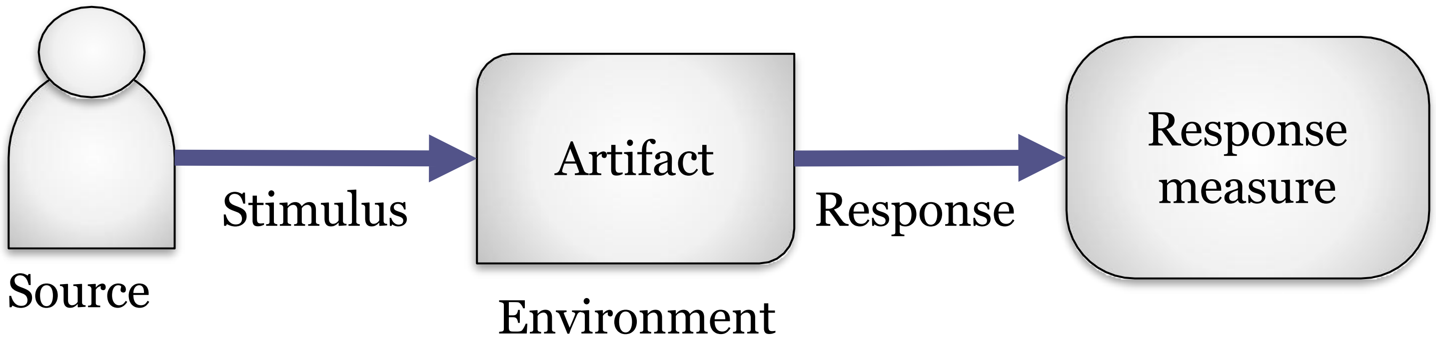
There is no single *perfect* system or architecture!

Specifying quality attributes

- 2 considerations:
  - QAs by themselves are not enough
    - They are non-operational
      - Example: It is meaningless to say that a system shall be modifiable or
        - maintainable
  - The vocabulary describing QAs varies widely
    - It is necessary to describe each attribute separately
- QA scenarios: A common form to specify QA requirements

Quality attribute scenario

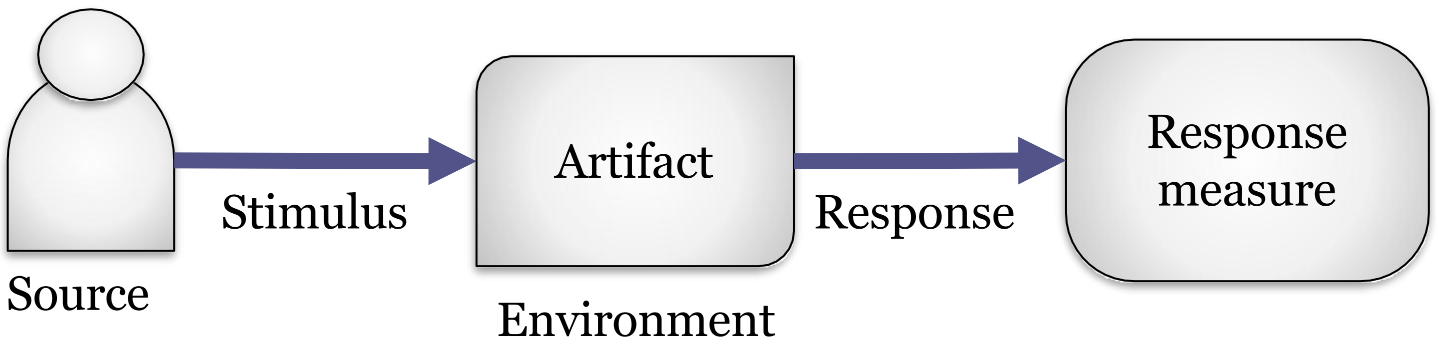
- Describe a stimulus of the system and a measurable response to the stimulus
  - Stimulus = event triggered by a person or system
  - The stimulus generates a response
- Must be testable
  - Response must be externally visible and measurable



Components of QA scenario

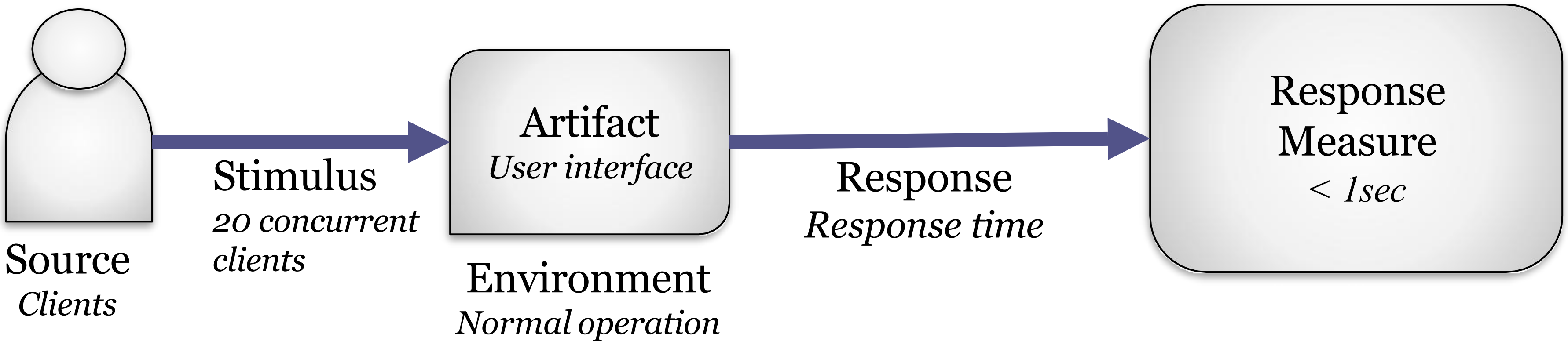
Source: Person or system that initiates the stimulus
Stimulus: Events that requires the system to respond
Artifact: Part of the system or the whole system
Response: Externally visible action
Response measure: Success criteria for the scenario
- Should be specific and measurable

Environment: Operational circumstances
- Should be always defined (even if it is "normal")



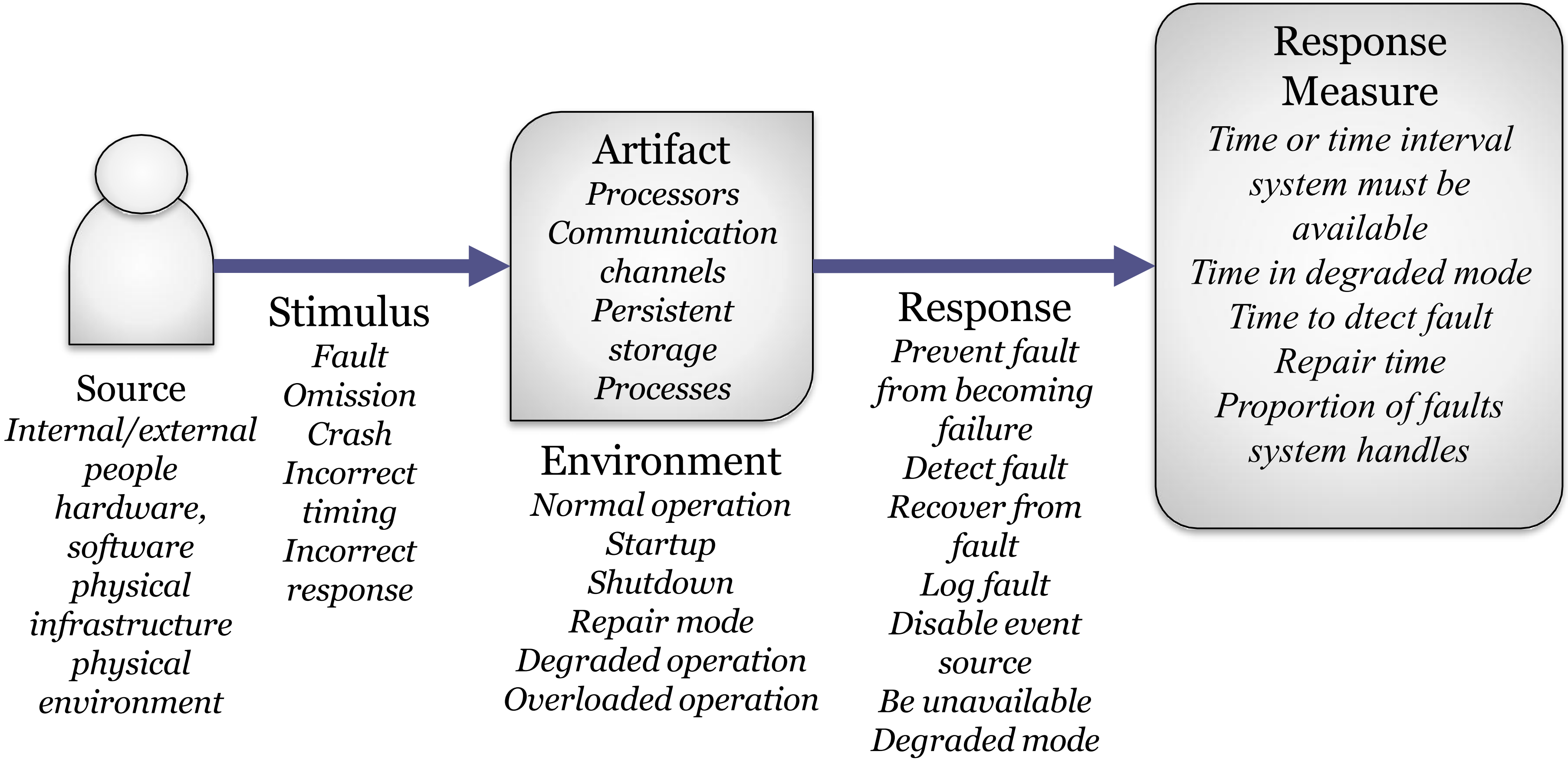
QA scenario, example 1

***Performance**: If there are 20 concurrent clients, the response time should be less than 1 sec. under normal operation*



Source of stimulus	Stimulus	Artifact	Environment	Response	Response measure
Clients	20 concurrent clients	User interface	Normal operation	Response time	<1sec
. . .	. . .	. . .	. . .	. . .	. . .

QA Scenario structure for availability



Types of QA scenarios

- Usage scenarios
  - The system is used (any use case or system function is executed)
    - Describe how the system behaves in these cases
      - Runtime, performance, memory consumption, throughput,...
- Change (or modification) scenarios
  - Any component within the system, its environment or its operational infrastructure changes
- Failure scenarios:
  - Some part of the system, infrastructure or neighbours fail

Prioritizing QA scenarios

- Scenarios should be prioritized
  - 2 values (Low/Medium/High)
    - How important it is for success (ranked by customer)
    - How difficult it is to achieve (ranked by architect)

Ref	Quality Attribute	Scenario	Priority
1	Availability	When the database does not respond, the system should log the fault and respond with stale data during 3 seconds	High, High
2	Availability	A user searches for elements of type X and receives a list of Xs 99% of the time on average over the course of the year	High, Medium
3	Scalability	New servers can be added during the planned maintainance window (less than 7 hours)	Low, Low
4	Performance	A user sees search results within 5 seconds when the system is at an average load of 2 searches per second	High, High
5	Reliability	Updates to external elements of type X should be reflected on the application within 24 hours of the change	Low, Medium
	. . .	. . .	. . .

Identifying quality attributes

- Finding QAs
  - Most of the time, QAs are not explicit
    - They're only verbally said alongside func. requirements
    - Usually implicit or said without much thought
  - Software architect must do educated guesses
- Quality Attribute Workshops
  - Meetings where stakeholders specify QAs
- Formal checklists
  - ISO25010
  - Wikipedia: https://en.wikipedia.org/wiki/List_of_system_quality_attributes

Typical quality attributes

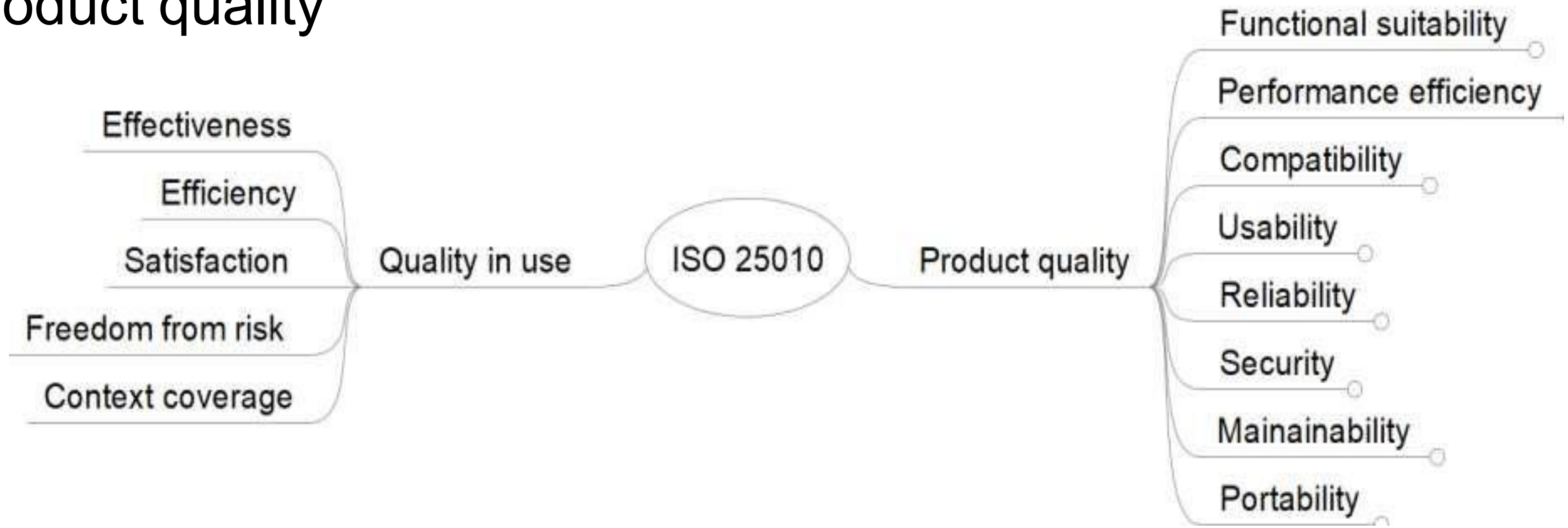
Availability
Modifiability
Performance
Security
Testability
Maintainability
Usability
Scalability
Portability
Changeability
. . .

ISO-25010 Software Quality Model

Systems and software Quality Requirements and Evaluation (SQuaRE)

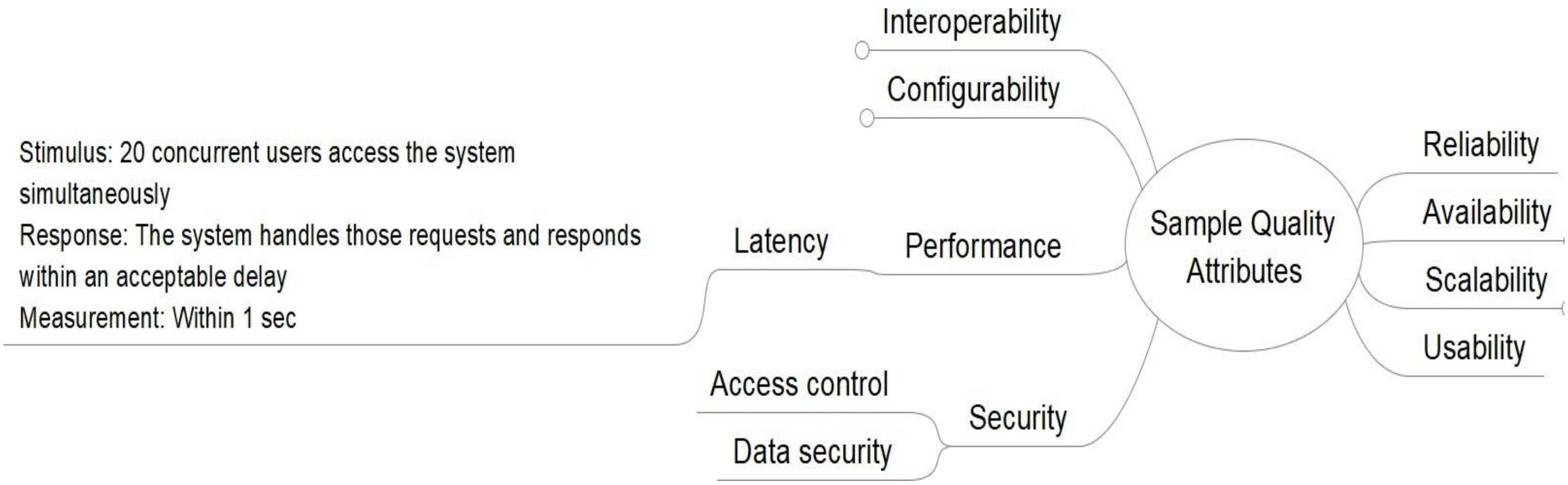
2 parts:

- Quality in-use
- Product quality



Quality Attribute tree

Mindmap representations can be useful to visualize QA scenarios



Measuring quality attributes

QA scenarios require response measures

Common problems:

- Many QA have vague meanings
  - Example: deployability, scalability
- Wildly varying definitions
  - No universal definitions
- Too composite
  - Usually, a QA is composed of several features

Software metrics

- Goal: Objective measures which can be automatically computed

  Numerous metrics for different aspects

- More information:
  - https://en.wikipedia.org/wiki/Software_metric

Operational measures

Lots of metrics

Absolute values

Examples: Number of users, number of errors

Statistical values and models

Example

$$availability = \frac{MTBF}{MTBF + MTTR} \text{ where } \begin{array}{c} MTBF = mean\ time\ between\ failures\ (uptime) \\ MTTR = mean\ time\ to\ recover\ (downtime) \end{array}$$

Availability	Downtime/90days	Downtime/year
99,0%	21hours, 36 minutes	3 days, 15.6 hours
99,99%	12 minutes, 58secs	52min, 34 secs
99,9999%	8 secs	32 secs

Structural measures

Measure the structure of the modules

Examples:

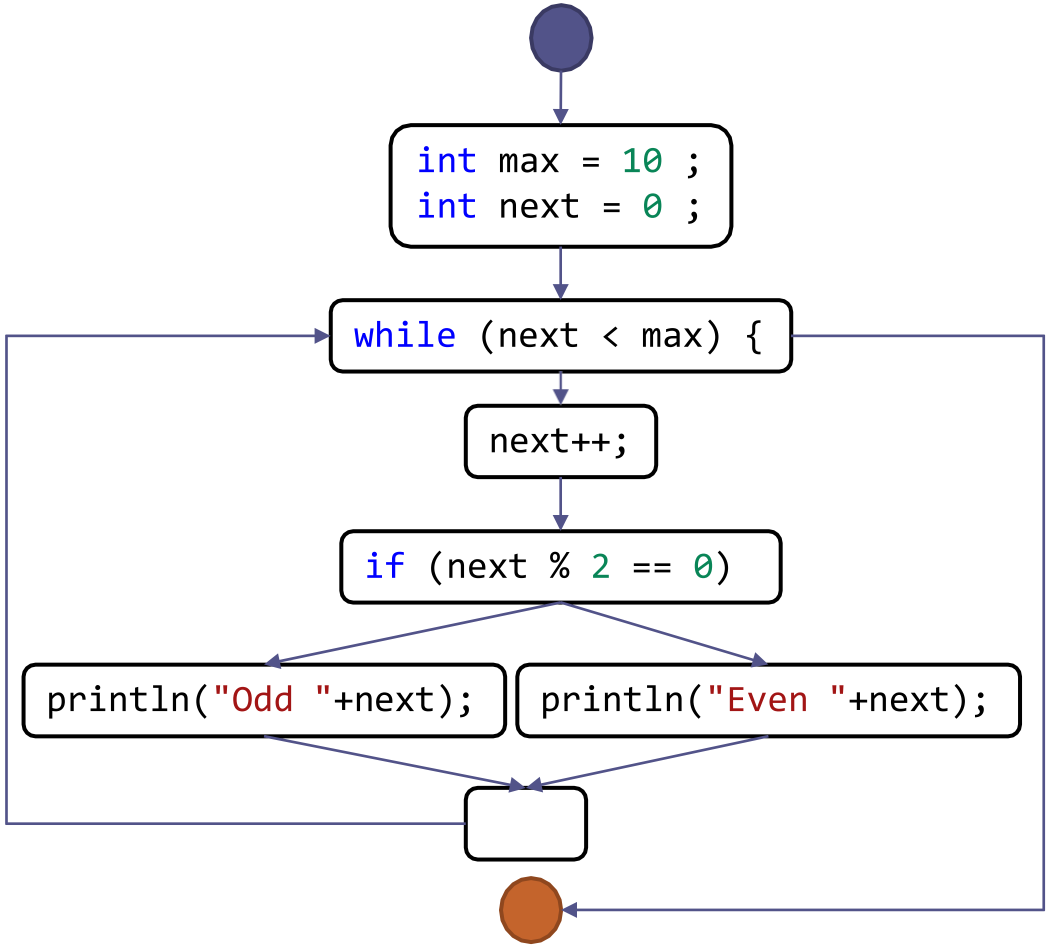
- Cyclomatic complexity (CC) from McCabe, 1976
  - Goal: Objective measure of complexity of code
  - It is obtained from the program control flow graph

$CC = E - N + 2$ where $E = number\ of\ edges$, $N = number\ of\ nodes$

Example:

```
public static void main(String[] args) {
    int max = 10 ;
    int next = 0 ;
    while (next < max) {
      next++;
      if (next % 2 == 0)
      System.out.println("Even " + next);
      else
       System.out.println("Odd " + next);
     }
   }
```

$CC = 10 - 9 + 2 = 3$



Process measures

- Measure process aspects like agility, testability, maintainability, etc.

Examples:

- Testability: Code-coverage during testing
  - Percentage of lines of code that have been run during the test phase
- Deployability:
  - % of successful to failed deploys
    How long deployment takes
    Issues/bugs in deployment
  - ...

Governing quality attributes

- Evolutionary architectures
  - Fitness functions: mechanism that provides objective integrity assessment of some combination of quality attributes
- Make QA measures and evolve the architecture